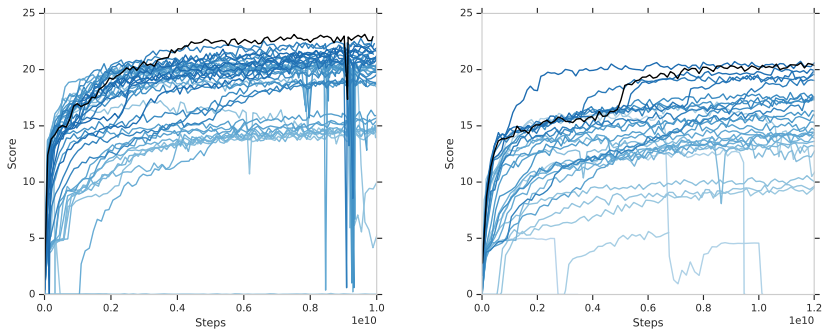


Figure 3: ACHA results without evolution, using 50 independent runs for two players (left), and 30 independent runs for four players (right).

hints to indicate which card of the teammate can be played. Different agents also use the rank-hints to indicate playability of cards in different slots. Details examining specific examples of learned policies are in Appendix A.2.

In contrast to ACHA, the Rainbow agents exhibit low variance across different independent training runs, as shown by the learning curves in Figure 4. In this case each line represents an independent training trial. In addition, Rainbow agents tend to converge to similar strategies, seemingly identifying the same local minima. In particular, Rainbow agents are 3-4 times less likely to hint for color than ACHA, and when they do there is no evidence of specific conventions associated with the color hinted. Instead all Rainbow agents we looked at primarily hint for rank, typically indicating that the most recent card is playable. See Appendix A.2 for details. We speculate that two factors contribute to this consistency across different runs. First, Rainbow has a one-step memory for the past action and no memory of past observations. This drastically reduces the space of possible strategies. Second, Rainbow is a value-based method and starts out with a high exploration rate. During the initial exploration, since agents fail to successfully finish any games without running out of lives, Q-values will tend towards zero. This might cause agents to learn from effectively the same starting Q-values, even across independent runs.

4.4. Experimental Results: Ad-Hoc Team Play

The policies learned by the aforementioned ACHA, BAD and Rainbow agents are moderately effective in self-play. We now investigate these agents in the ad-hoc team setting, where teammates are using different conventions. In particular, we examine the performance of different ad-hoc teams of ACHA and Rainbow agents. Since we established in Section 4.3 that Rainbow agents learn nearly identical strategies across different runs, for the rest of the section we consider a population of agents made up of the best performing Rainbow agent and a pool of independently trained ACHA agents taken from the top ten agents according to final training-time performance from Figure 3.

We do not use the hand-coded agents in the ad-hoc setting, for a mix of practical and technical reasons. As written, none of the hand-coded agents interface with the learning environment or are even written in the same language, with a different non-trivial (and possibly error-prone) translation required to have the learning agents play

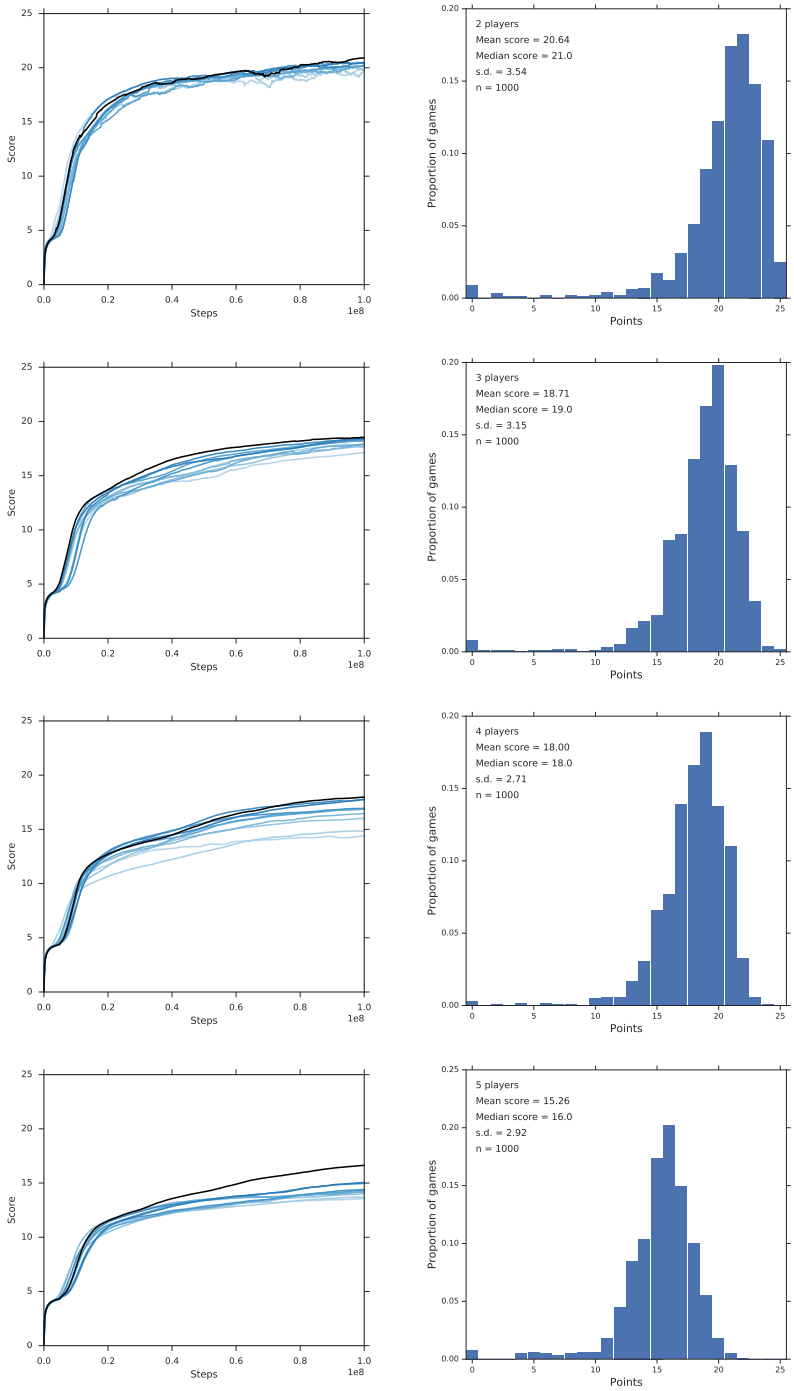
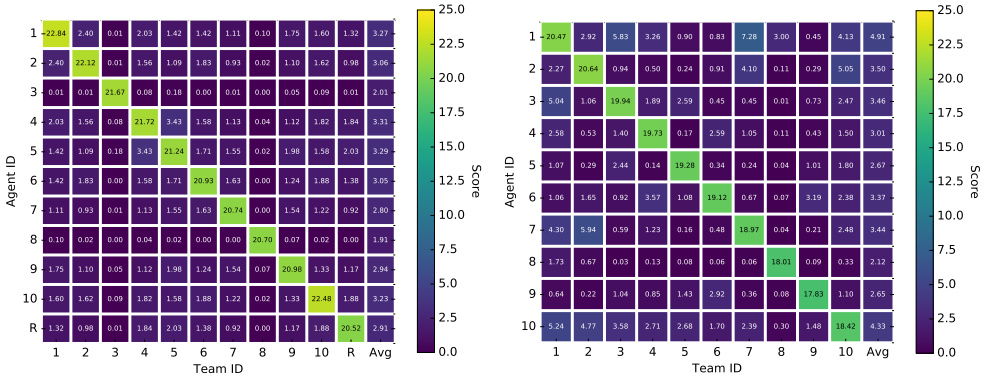


Figure 4: Rainbow results for two to five players, from top to bottom respectively. Performance curves (left) are training-time results from the current policy. Average scores and distributions (right) are test-time results from 1000 episodes generated using the agent with the best training score. ϵ in ϵ-greedy for all agents is set to zero.



(a) Ad-hoc results for two players.

(b) Ad-hoc results for four players.

Figure 5: Ad-hoc team results. Teams were constructed using the 10 best independently trained ACHA agents (**1**–**10**; see Figure 3) and the best Rainbow agent (**R**; see Figure 4). Mean scores over 1000 trials are shown. Standard error of the mean was less than 0.09 and 0.255 for two and four players, respectively.

with each hand-coded agent. We made an early effort to play games with combinations of ACHA agents and SmartBot, but these games always ended with SmartBot crashing because ACHA makes moves SmartBot assumes should never be made. Simple fixes only led to a score of zero points. Given the effort to connect different agents, and the complicated action coding of agents like HatBot and WTFWThat, we assume without further evidence that other combinations of hand-coded agents would have equally bad performance.

As neither of ACHA and Rainbow make use of sample play of their ad-hoc teammates, the ad-hoc team's performance will simply depend on the compatibility of the different protocols. The purpose of this section, then, is primarily to illustrate the difficulty in ad-hoc team play, and suggest a source for creating a diverse pool of agents for future evaluation.

Two Players. Figure 5a shows a crosstable of the agents' test performance in the ad-hoc setting. The entry for the i-th row and j-th column shows the mean performance of agent i playing in an ad-hoc team consisting of agent j, evaluated for 1000 games with a random player starting each game. The scores of 20 or greater along the diagonal entries show that the agents indeed perform well in self-play. However, when paired with other agents, performance drops off sharply, with some agents scoring essentially zero in any of these ad-hoc teams.

Four Players. We observe analogous results when evaluating ad-hoc teams in the four player setting. We used the top ten ACHA agents from Figure 3. Similar to the two player ad-hoc results, the entry for the i-th row and j-th column of Figure 5b shows the mean performance of agent i when playing with an ad-hoc team consisting of three other agent j players, evaluated for 1000 games with a random player starting each game. As in the two player results, the agents fare relatively well in self-play but performance dramatically decreases once we introduce a second unique agent to the team.

Developing agents that can learn from, adapt, and play well with unknown teammates represents a formidable challenge.